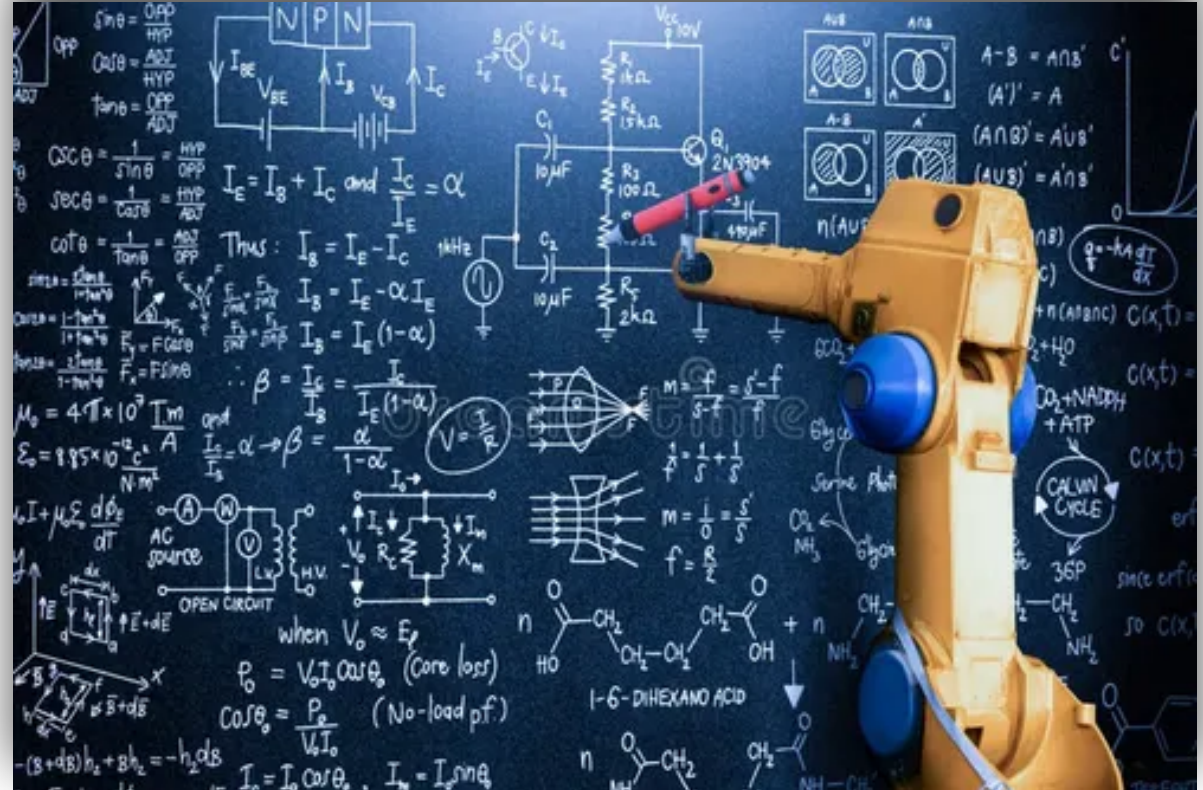


WEEK 2 – LECTURE 4

Artificial Intelligence Foundations

Performance Measures & AI Problem-Solving Approaches



Quick Recap: Problem Formulation



"If there are multiple solutions, how does AI decide which one is best?"

Motivation: Which Option to Choose?



Google Maps

Suggests two different routes to your destination.



Robot

Finds two different paths to pick up an object.



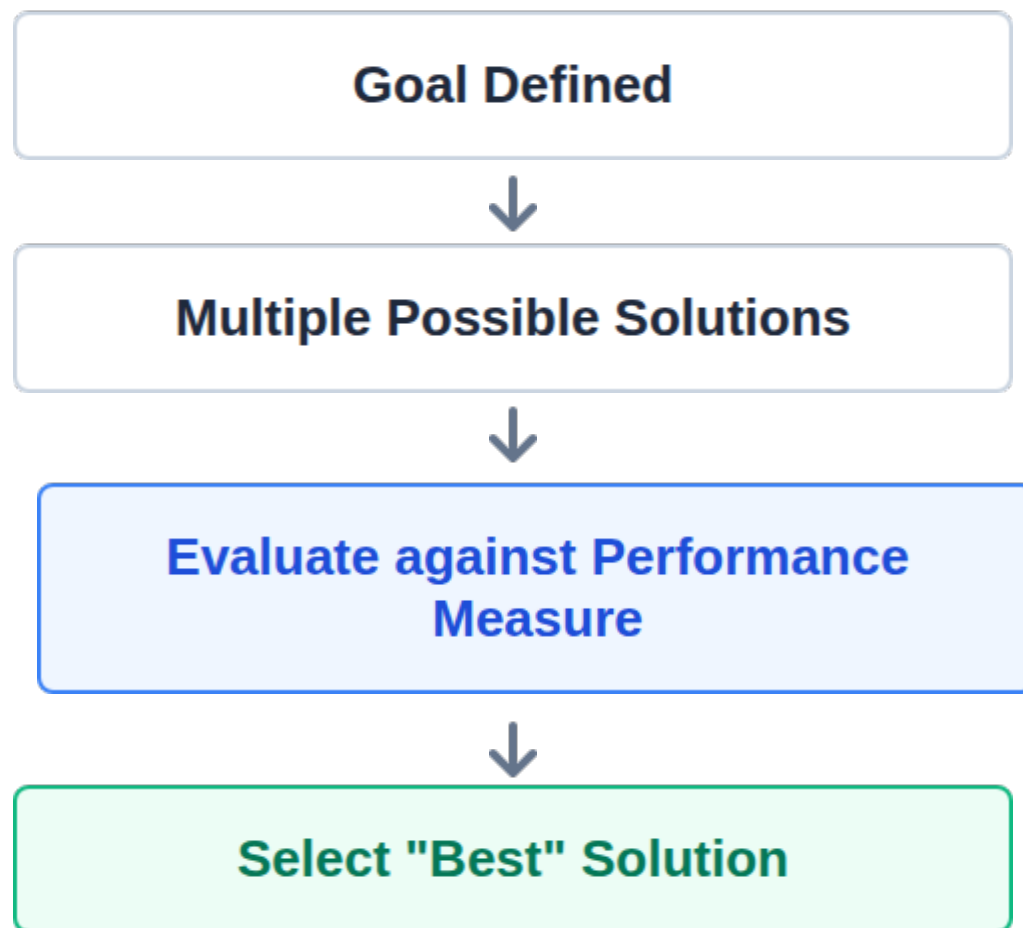
Chess AI

Has multiple legal moves available on the board.

How should the AI choose?

What is a Performance Measure?

The criterion used to evaluate how well an AI system achieves its goal.



Everyday Analogy: Food Delivery



A delivery agent needs to reach you.

They can optimize for different things:

-  Fastest delivery time
-  Shortest travel distance
-  Lowest fuel consumption
-  Highest customer satisfaction

Different objectives lead to different "best" routes.

Common Performance Measures in AI



Accuracy

Correctness of the output.



Speed

Time taken to find solution.



Cost

Financial or computational cost.



Safety

Minimizing risk or harm.



Resource Usage

Memory and power consumed.



Reliability

Consistency over time.

Real-World Examples

 AI System

 Primary Performance Measure

Google Maps

Shortest travel time

Spam Filter

Correct email classification (Accuracy)

Self-Driving Car

Safety (collision avoidance)

Netflix

Recommendation quality (User engagement)

Chess AI

Winning the game

Medical AI

Diagnosis accuracy



"If you designed an AI for a hospital emergency room, what would its most important performance measure be?"

🎯 Accuracy? ⌚ Speed? 🛡️ Safety?

Trade-offs in Performance

Robot Vacuum A



- ✓ Cleans very quickly.
- ✗ Misses many dusty areas.

VS

Robot Vacuum B

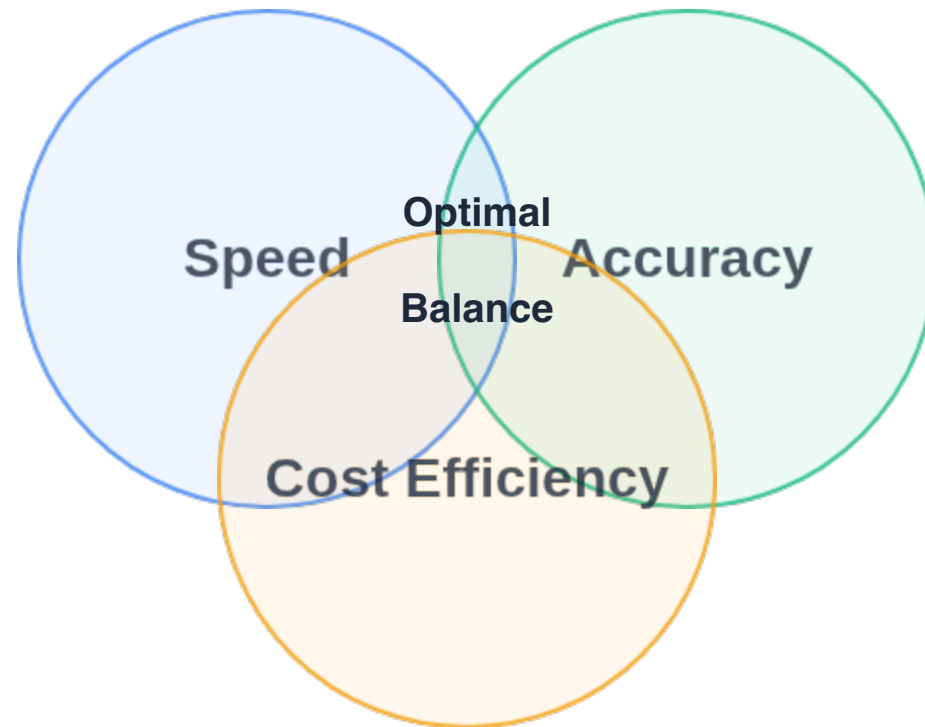


- ✓ Cleans thoroughly.
- ✗ Takes slightly longer.

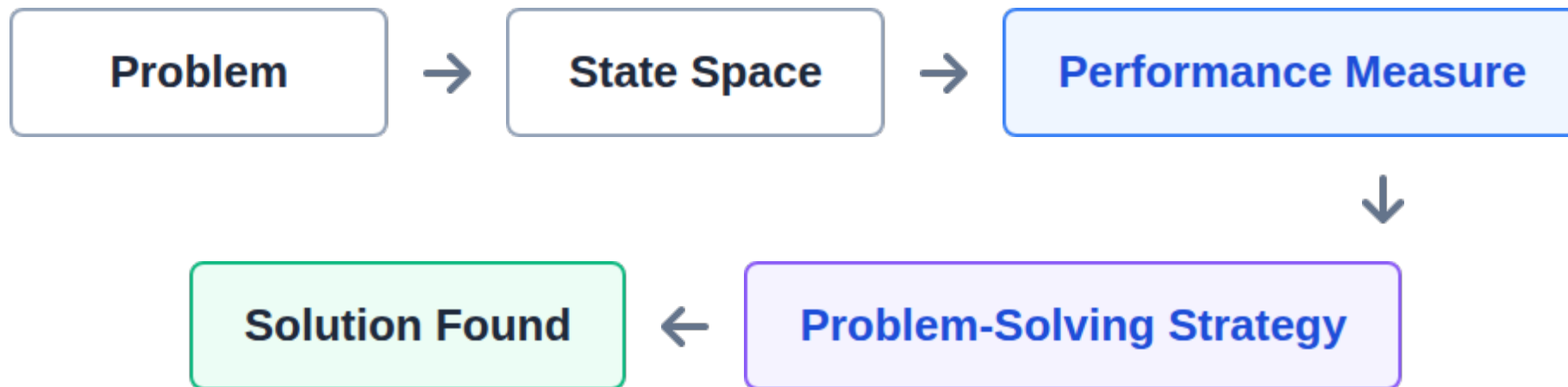
Which robot performs "better"?

Balancing Multiple Objectives

AI systems rarely optimize only one metric.



From Formulation to Solving





What is an AI Problem-Solving Approach?

After defining a problem and deciding how success is measured, AI must choose an appropriate strategy or method to actually find the solution.

Overview of Problem-Solving Approaches



Search-Based

Systematically exploring paths (e.g., Mazes, Navigation).



Knowledge-Based

Using rules and logic (e.g., Medical diagnosis).



Learning-Based

Improving from data (e.g., Image recognition).



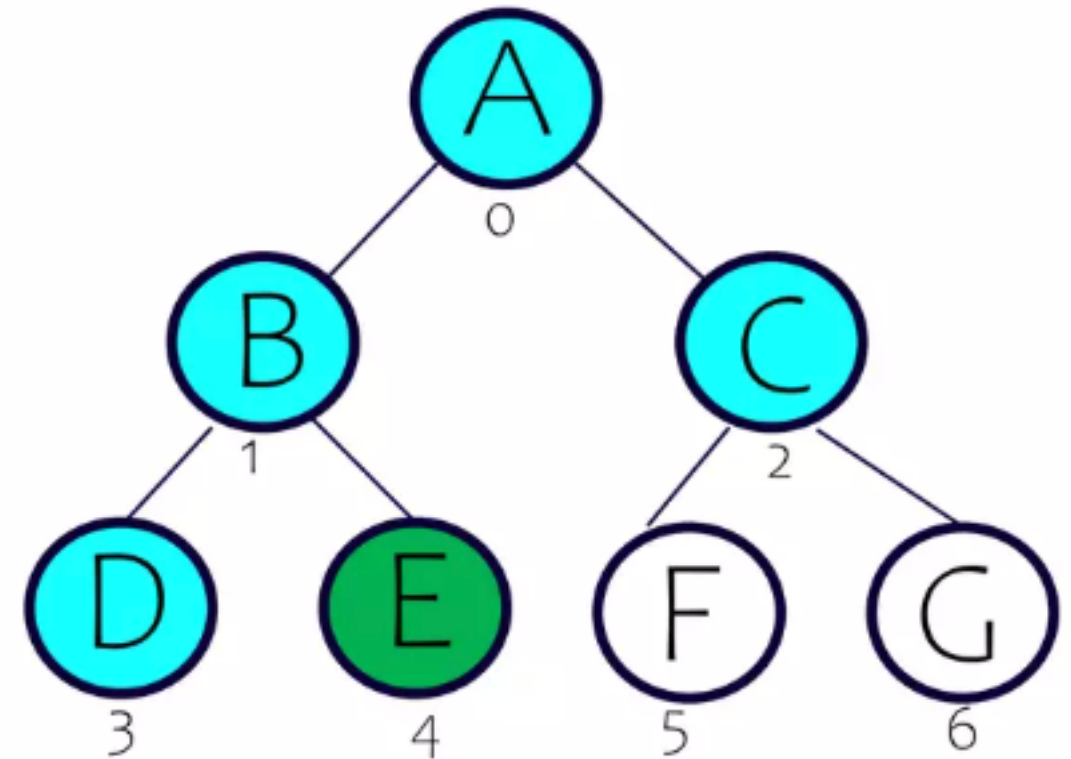
Optimization

Finding the absolute best state (e.g., Resource allocation).

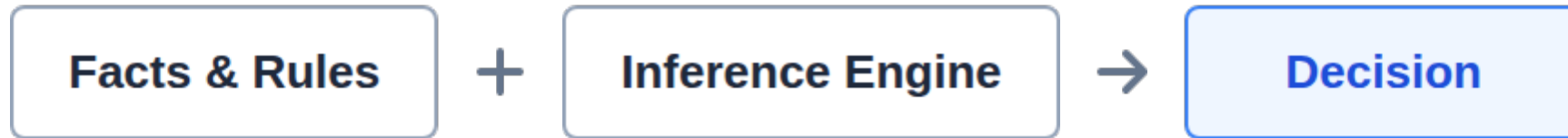
Search-Based Methods (Preview)

- 🔗 Systematically explores possible sequences of actions.
- 🌲 Builds a "search tree" of possibilities.
- ▶ Used when the sequence of steps matters.

* We will focus heavily on Search Algorithms starting next week.



Knowledge-Based Methods



Medical Expert Systems



Tax Rule Systems



Troubleshooting Assistants

Learning-Based Methods



Face Recognition

Movie Recommendations

Voice Assistants

Case Study: Google Maps Route Selection

Road Network



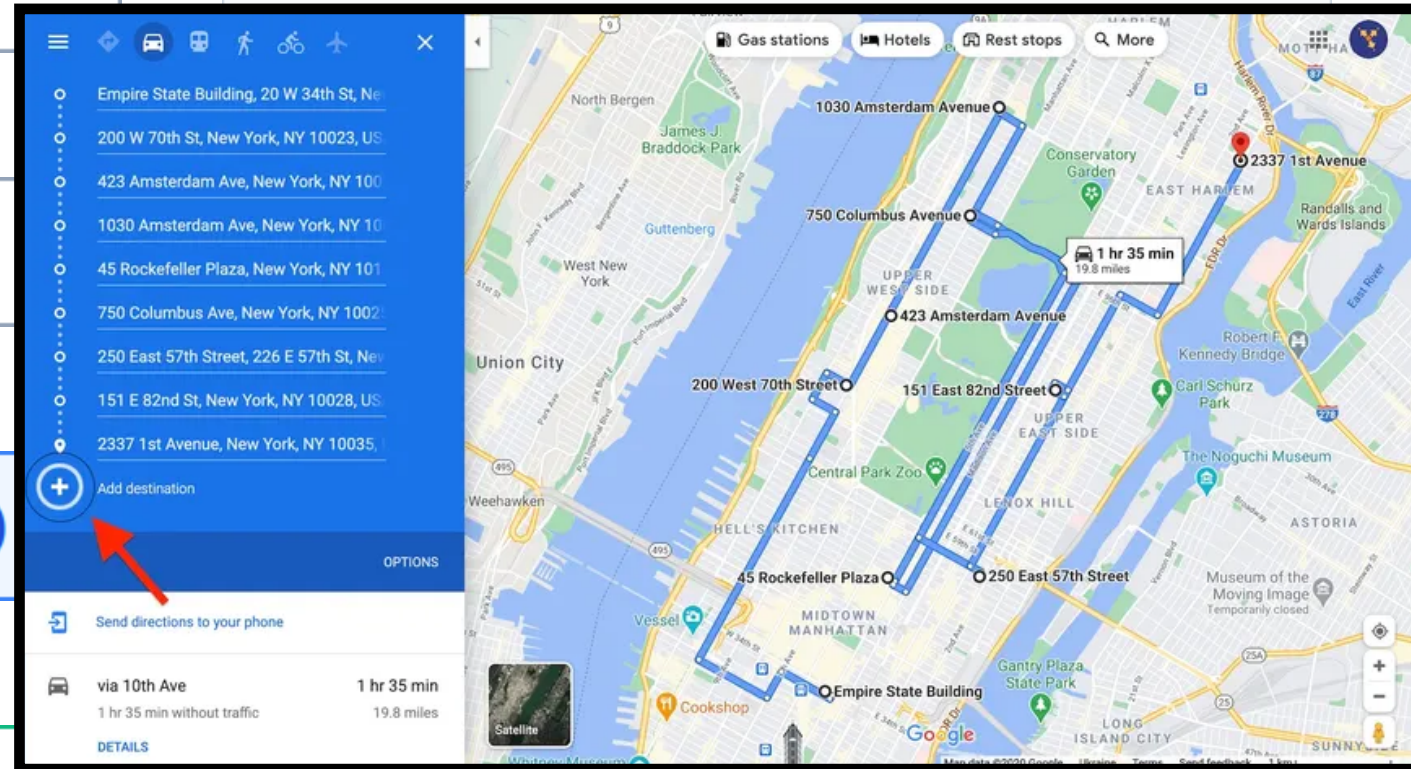
Find Possible Routes (Search)



Evaluate Travel Time (Performance)



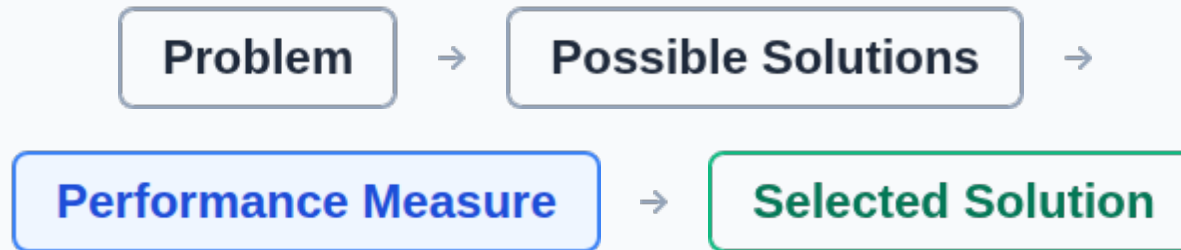
Choose Best Route





Board Activity

Draw:



**Illustrating why different criteria produce different "best" paths on a simple graph.*

Common Misconceptions



"Fastest always means best."

Not if safety or accuracy is more critical.



"The shortest path is always optimal."

It might be heavily congested (traffic) or cost more.



"AI always finds one perfect solution."

Often, it finds a "good enough" solution based on trade-offs.



"Every problem uses the same performance measure."

The measure must be customized to the specific problem.

Key Takeaways



Performance
measures define
success.



Different systems
optimize different
objectives.



AI often balances
multiple goals.



Strategy depends on
the problem.



Search is one of
many approaches.

Quick Quiz

Which metric is most critical for a self-driving car?

- ☐ A) Arriving as fast as possible
- ☐ B) Using the least amount of fuel
- ☐ C) Passenger and pedestrian safety
- ☐ D) Finding the most scenic route

Quick Quiz

Which metric is most critical for a self-driving car?

- ☐ A) Arriving as fast as possible
- ☐ B) Using the least amount of fuel
- ☒ **C) Passenger and pedestrian safety**
- ☐ D) Finding the most scenic route

Preview: Unit II - Search Algorithms

Coming Up Next Week:

🔍 How does AI actually search?

📖 Breadth-First Search (BFS)

⬇️ Depth-First Search (DFS)

Get ready to dive into the algorithms!



VS



Image Sources

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